Energy Stock Price Forecasting Using LSTM

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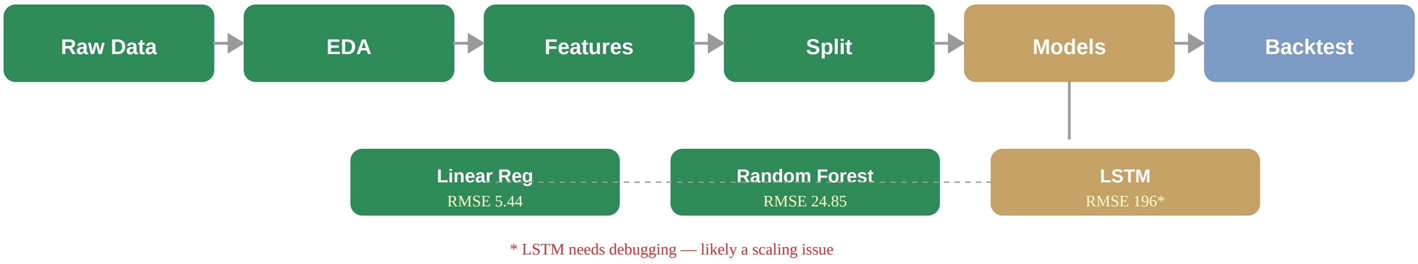
Stock	Data	Features	Models	Framework
NTPC.NS	5yr daily OHLCV	17 technical indicators	LR + RF + LSTM	PyTorch + sklearn

Objective

Forecast next-day closing prices of NTPC (India's largest power utility) using an LSTM neural network, benchmarked against Linear Regression and Random Forest baselines. Complete pipeline: data collection → EDA → feature engineering → model training → backtesting.

Pipeline



Features (17 Technical Indicators)

Category	Indicators	Purpose
Trend	SMA(20/50/200), EMA(12/26), MACD	Market direction & momentum
Momentum	RSI(14), Stochastic K	Overbought/oversold detection
Volatility	Bollinger Bands, ATR	Price spread & range
Volume	OBV, Volume Ratio	Trend confirmation via volume
Custom	Returns(1d/5d/20d), Price Range	Multi-horizon returns

Model Results

Model	RMSE	MAE	Dir. Accuracy	Notes
Linear Regression	5.44	4.06	51.74%	Best — features are linearly related to target
Random Forest	24.85	22.30	49.57%	Step-wise splits can't fit smooth price curves
LSTM (2-layer, 64h)	196.47	194.93	48.00%	Needs debugging (likely scaling/target issue)

Key insight: Linear Regression won because top features (SMA_50, SMA_200, EMA) are smoothed prices — the feature→target relationship is inherently linear. RF rounds off at tree splits; LSTM likely has a target scaling mismatch that will be fixed in Week 2.

LSTM Architecture

Layer	Config
Input	17 features × 30 timesteps
LSTM ×2	64 hidden, dropout=0.2
Dense	64→32 (ReLU) → 32→1
Training	Adam, LR=0.001, early stop (patience=10), grad clip=1.0

3-Week Project Timeline (Aug 1–21)

Week	Days	Tasks	Status
1 Foundation	Aug 1-2	Setup, download data, EDA (6 plots), feature eng., baselines, LSTM	**DONE**
	Aug 3-4	Backtesting (Sharpe, drawdown), debug LSTM, first GitHub push	**TODO**
	Aug 5-7	Code review, add comments, understand model tradeoffs	**TODO**
2 Polish	Aug 8-9	Write README.md with results + plots, update report PDF	**TODO**
	Aug 10-11	Tune LSTM hyperparams, try more stocks (Reliance, ONGC)	**TODO**
	Aug 12-14	Study LSTM internals, Sharpe derivation, interview prep	**TODO**
3 Extend	Aug 15-17	Optional: crude oil feature, GRU/Transformer comparison	**TODO**
	Aug 18-19	Final cleanup, docstrings, pin repo on GitHub, update resume	**TODO**
	Aug 20-21	Mock interview: explain project end-to-end in 5 min	**TODO**

Progress Tracker

Milestone	Status	Done
Data collection + EDA (6 plots, ADF test)	**Complete**	100%
Feature engineering (17 indicators)	**Complete**	100%
Baseline models (Linear Reg + Random Forest)	**Complete**	100%
LSTM model (training done, needs debug)	**Partial**	80%
Backtesting + trading strategy	**Pending**	0%
README + documentation	**Pending**	0%
Resume + interview prep	**Pending**	0%

Technical Skills

Domain	Tools & Concepts
Data	yfinance API, pandas, pathlib, CSV pipelines
ML	scikit-learn (LinearRegression, RandomForest), StandardScaler, chronological split
Deep Learning	PyTorch LSTM, DataLoader, early stopping, LR scheduling, gradient clipping
Visualization	matplotlib, seaborn (histplot, heatmap), statsmodels (ACF/PACF)
Statistics	ADF test, stationarity, skewness, kurtosis, max drawdown
Quant Finance	Sharpe ratio, backtesting, long-only strategy, directional accuracy
DevOps	Git, GitHub, .gitignore, virtual environments

Next steps: Debug LSTM scaling issue → backtesting → README → tune hyperparams → add more stocks → GRU/Transformer comparison → interview prep.

GitHub: github.com/aariiparekh3012-collab/project1-aarya